

Avi Gobrin

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EDUCATION

University of Toronto

2023 – Expected Apr 2027

Bachelor of Applied Mathematics Specialist, Statistics Major, Computer-Science Minor

Toronto, ON

Relevant Coursework

Real Analysis, Advanced Calculus, Nonlinear Optimization, Probability and Statistics I/II, Multivariate Data Analysis, Algorithm Design and Analysis, Graph Theory, Combinatorics

EXPERIENCE

Nextdoor

May 2026 – Present

Machine Learning Intern

San Francisco, CA

- Designed and deployed a quantitative model that ranks neighborhoods by family-friendliness, turning an open-ended research question into an interpretable scoring system live in production for users.
- Built the signal pipeline ingesting external APIs and internal data through Databricks and SQL over tens of thousands of records, engineering noisy inputs into a single weighted index and comparing multiple model families before selecting the production scorer.
- Owned the system end to end, from data processing to the backend scoring service and the frontend that surfaces ranked recommendations, and resolved bugs across the pipeline and frontend.

Shopify

May–Aug 2025

Software Engineering Intern

Toronto, ON

- Increased new-merchant activation and retention, routing roughly 100 merchants/day through guided onboarding, by launching a partner-attach system that automatically pairs each new merchant with a vetted onboarding advisor.
- Brought German VAT submissions into compliance with updated tax regulations, unblocking hundreds of affected merchants and reducing compliance risk, by integrating the German tax-code API and aligning Shopify's existing tax flows with the new legal requirements.
- Improved system reliability and reduced recurring issues by resolving cross-disciplinary technical debt assigned by adjacent teams, collaborating with engineering and product stakeholders to ship scalable fixes.

PROJECTS

Multivariate Analysis of Student Outcomes (OULAD), Independent Research [\[Report\]](#)

2026

- Recovered latent engagement and performance structure across 23,301 student-course records via a 3-component PCA basis (72.9% variance) and varimax factor model, after formally rejecting multivariate normality (Mardia, Box's M).
- Showed PCA, factor analysis, GMM, and LDA converge on one two-dimensional structure, and benchmarked classifiers under 5-fold CV (logistic regression 67.0% vs. 53.0% baseline) with robustness checks on covariance assumptions.

Autonomous Planning & School Knowledge Agents, Personal Project

2026

- Built an autonomous daily-planning agent (Claude Code, Obsidian) that reviews each day's completed work, writes structured notes, and regenerates a time-blocked calendar from a parsed task file and traffic-light priority scheme, cutting daily planning from about 30 minutes to under 2 minutes and saving an estimated 3–4 hours per week.
- Built a companion school-workflow agent that auto-formats lecture and problem-set notes into a cross-linked knowledge base of theorem references, cutting note-cleanup and material-lookup time an estimated 40% to accelerate homework and exam preparation.

F1 Race Winner Predictor, Personal Project

2024

- Built a supervised ML pipeline predicting Formula 1 race winners, engineering features from driver/constructor rolling form, qualifying results, driver–track history, pit-stop reliability, and weather over historical race data.
- Evaluated on held-out races, the classifier reached 40% race-winner and 60% podium-finish accuracy.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, JavaScript, Bash

Libraries & Frameworks: NumPy, Pandas, SciPy, scikit-learn, statsmodels, Matplotlib

Tools & Platforms: Jupyter, Databricks, Git, GitHub, Excel, L^AT_EX

AI Tools: Claude, Claude Code, Cursor, Hermes, LLM Agents